AI-Based Candidate Recommendation for Company Vacancies: Emphasizing Soft Skills and Competencies

Abstract—**Historically, the recruitment process relies predominantly on resumes, academic achievement, and technical testing, but these can overlook significant soft skills for workplace success. Soft skills such as confidence, communication, adaptability, and stress tolerance have direct influence on the performance of employees but are tested subjectively and non-systematically in an interview. This paper introduces a multimodal approach of an AI-recommended candidate system that objectively evaluates the soft skills of candidates through a multimodal approach. The framework integrates four modules: (i) speech and tone analysis for verbal confidence, (ii) hand movement recognition for confidence estimation, (iii) eye-contact detection for self-confidence and attentiveness, and (iv) facial expression analysis for stress. All the modules output binary decisions Confident/Non-Confident or Stress/Non-Stress, which are integrated to present a holistic perspective of the candidate. Minimum decision-making assessment is also offered to optimize evaluation. The application is implemented as a web app recording candidate interview, processing multimodal signals with machine learning models, and providing real-time recommendations to recruiters through an interactive dashboard. Experimental outcomes demonstrate superb accuracy from all modules and superb user acceptance, confirming the feasibility of applying multimodal AI to enhance fairness and richness in hiring. This study illustrates the potential of AI to reduce bias, increase transparency, and support data-driven decision-making that is more subtle than résumé-based assessment.**

Keywords: Artificial Intelligence, Candidate Recommendation, Soft Skills, Confidence Prediction, Stress Detection

I. INTRODUCTION

Conventional hiring has focused on resumes, academic performance, and technical examinations. While these are useful, they merely offer a one-sided view of a candidate's fit for a job. Modern organizations increasingly realize that behavioral skills, particularly confidence and stress management skills are essential to workplace success [3], [4], [9]. These abilities are typically assessed subjectively through interviews, where human judgment is prone to inconsistency and bias [7], [13], [22]. This lack of objectivity could lead to the selection of employees through uninformed decisions and reduced fairness in hiring.

Artificial intelligence (AI) developments can eliminate these weaknesses by providing objective and fact-based analysis of the candidates' responses. With multimodal learning and computer vision, AI can capture both verbal and non-verbal cues and gauge confidence and stress more effectively [10], [15]. For instance, speech changes such as pitch, tone, and rhythm are excellent confidence or hesitation indicators [11], [12], [14]. Apart from that, overt and conscious hand gestures are associated with assertiveness and confidence, while repetitive or defensive movements indicate nervousness [17],[19]. Sustained eye contact is a clear indication of attentiveness and confidence, while avoidance of eye contact could be an indication of discomfort or lack of confidence [1], [2]. In contrast, facial actions such as furrowed brows, pursed lips, or brief micro-expressions are widespread indicators of stress and emotional tension [5], [6], [8].

Although previous work on affective computing has illustrated the potential of such behavioral cues for inferring psychological and emotional states, most of the existing studies investigate a single modality only, thereby limiting strength and scalability [15], [20]. Few approaches blend multiple modalities into a unified framework that can both measure confidence and stress in recruitment scenarios. Furthermore, the problems of cultural differences in non-verbal signals, training data bias, and issues of fairness and transparency remain significant hurdles for real-world deployment [13], [16], [23].

This research bridges these gaps by proposing a candidate recommendation application based on AI that blends four behavioral dimensions into a multimodal framework. This calculates a confidence score by analyzing speech, hand movements and eye contact, and stress score from facial expressions. These are summarized into an understandable profile and delivered to recruiters through a web-enabled application. By highlighting confidence and stress, two of the most important predictors of job performance, the system reduces the reliance on personal views and increases the fairness, transparency and efficiency of recruitment decisions [21], [24].

II. LITERATURE REVIEW

It is no surprise that confidence and stress management are considered soft skills and are helpful in predicting performance in the workplace and long-term employability. Technical fit will invariably remain the shorthand recruiting consideration. However, it is becoming more apparent to employers that behavioral characteristics may shape an employee's ability to work as a team member, respond to their environment (e.g., adapt to change), and function under pressure ([3], [4], [9]). Certainly, soft skills will be important, if not vital to the role; nevertheless, conventional recruitment practices such as résumé screening and structured interviews have their limitations in incorporating these dimensions, and as such will not reflect candidates' full potential. Similarly, while development review can be useful, for the most part it involves the subjective judgment of the reviewer and so may be vulnerable to inconsistency and inadvertent bias ([7], [13], [22]). It follows that workplace evaluations of AI will become extensive research into this area, so that the aim is to provide an objective, data-driven appreciation of prospective employee behavior.

Among the behavioral cues studied, speech and tone have emerged as strong predictors of confidence. Variations in pitch, intonation, rhythm, and vocal energy provide measurable indicators of self-assurance or hesitation [11], [12]. With the advent of deep learning, architectures such as convolutional and recurrent neural networks have been successfully applied to analyze prosodic features, enabling accurate classification of confident versus hesitant speech [14]. These developments demonstrate the value of verbal cues as consistent markers of confidence, particularly when supported by open-source feature extraction tools such as openSMILE [12].

Non-verbal communication has also been shown to significantly contribute to confidence perceptions. Research suggests that deliberate and open gestures in hand and arm motion are often indicators of assertiveness, whereas nervousness is often represented by repetitive gestures or defensive gestures [17][19]. With advances in computer vision, these gestures can be captured and classified in real time. Existing frameworks like MediaPipe and computer vision models using convolutional neural networks (CNN) can classify hand gestures reliably [17], [18]. Furthermore, eye contact has been known to be an indicator of attentiveness and confidence. Deeper engagement is reflected in sustained gaze and gazing away or not maintaining eye contact often signals discomfort or lack of confidence [1], [2]. New computer vision and eye tracking solutions can capture significant metrics such as fixation durations and blink rates are now possible. This allows one's eye movement to be quantified to measure confidence behavior objectively.

Although confidence is primarily replicated in speech, gestures, and gaze, stress is typically replicated through facial expressions. Darwin [5] and Ekman and Friesen [6] explained that humans display and carry their emotions via micro-expressions universally, and subsequent research has demonstrated that stress can be detected using facial cues. Deep learning techniques, such as Convolutional Neural Networks, have improved the ability to identify stress indicators such as furrowed brows, tightened lips, and tightened jaws, with a reasonably high level of accuracy in real world scenarios [8]. Overall, these results indicate that multimodal analysis provides a better or richer understanding of a candidate's behavior by analyzing both verbal and non-verbal cues, compared to one mode alone [10], [15], [20].

AI approaches for recruitment have made progress, challenges still exist. Existing literature generally discusses a single modality limiting the robustness and generalizability of the findings [10], [15]. It has also been noted there are few studies that combine these systems to predict confidence and the detection of stress into a single system that is useful in hiring workflow. Also, using AI in recruitment raises ethical and cultural issues. Bias in the training dataset would result in ineffective recruitment potentially unhelping candidates from underrepresented groups, whilst cultural differences in non-verbal behaviors can lead to adverse misinterpretation of gestures, gaze, or facial expressions [13], [16], [23]. Researchers believe these systems should have fairness, transparency and explain ability at the center of the design of recruiting tool to build trust of the clients and candidates [16].

Finally, there are important practical considerations that limit the use of recruiting tools such as Integrating with HR tools, immediate performance, and interpretabilities to allow to deduce how improvements can be made for their organizations [21], [24].

Features / Gaps	[11], [12], [14]	[17]–[19]	[1], [2]	[5], [6], [8]	[10], [15], [20]	Proposed application
Speech & tone analysis for confidence prediction	Yes	No	No	No	Partial	Yes
Hand gesture analysis as indicator of confidence	No	Yes	No	No	Partial	Yes
Eye contact / gaze tracking for confidence assessment	No	No	Yes	No	Partial	Yes
Facial expression recognition for stress detection	No	No	No	Yes	Partial	Yes
Multimodal integration of behavioral cues (speech, gesture, eye, face)	No	No	No	No	Yes (limited)	Yes (complete)
Unified framework for confidence & stress evaluation in recruitment	No	No	No	No	No	Yes
Real-time recommendation system for candidate profiling	No	No	No	No	No	Yes

Table1: Gap in literature

III. Methodology

The proposed AI-based candidate recommendation approach processes two interview-determining behavioral traits, confidence and stress using a multimodal deep learning framework. As shown in Figure 1, the approach consists of four expert subsystems: (1) speech, (2) facial expression, (3) eye engagement, and (4) hand gesture analysis. The input modality of each subsystem is processed independently, features are extracted, and inference with a modality-optimized neural model is done. The outputs are normalized to 0–100 and afterwards aggregated into a role-aware candidate score through weighted aggregation. Inference in real-time is supported through a web-based application so that recruiters can rate candidates during interviews.

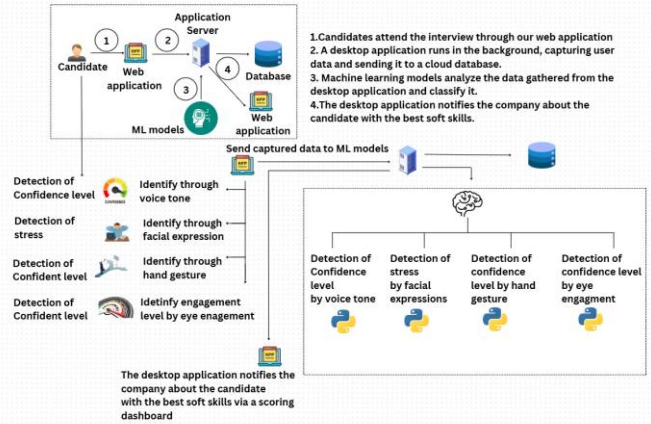

Figure 1: System Overview Diagram

A. Data Collection and Preprocessing

The system captures multimodal signals from candidates throughout interviews across four input modalities: speech, facial expressions, eye engagement, and hand gestures. Audio was recorded at 16 kHz to pick up vocal cues about confidence and stress. Facial expressions were recorded as RGB webcam streams at 25–30 frames per second (FPS) for micro-expression analysis about stress. Eye engagement was computed from cropped eye landmark detections, allowing measurement of gaze direction, fixation rate, and blink rate. Upper-body video streams were utilized to capture hand gestures, where a series of motion captured non-verbal behavior cues related to confidence.

Preprocessing was conducted specifically for every modality to prepare the data for feature extraction and model training. Resampling of speech recordings was conducted at 16 kHz, pre-emphasis filtering (0.97), segmenting into 25 ms frames with a hop of 10ms, and Hamming windowing. Mel-Frequency Cepstral Coefficients (MFCCs) with delta and delta-delta features were computed, then time-normalized (Z-score or min–max scaling). Facial video streams were subjected to face detection and alignment, cropping frames resized to 48×48 pixels, converted to grayscale, and normalized to the [0,1] space. For eye gaze, eye regions were cropped from facial landmarks, resized to 48×48 gray-scale images and normalized; temporal features such as blink rate and fixation stability were also computed. Hand gesture pre-processing included 2D hand point detection with models such as MediaPipe/OpenPose, creating motion sequences for temporal processing. In CNN-based processing, upper-body and hand images cropped were resized and normalized. Finally, all modalities were translated into one-hot labels for learning and temporal aggregation was used to determine trial-level scores from frame-wise predictions.

B. Feature Extraction and Model Training

All modalities employed bespoke feature extraction and deep learning models. Speech utilized MFCC-based spectrograms that were passed through a convolutional neural network (CNN) to predict emotional states, which were then mapped to confidence and stress levels. Facial expressions were analyzed via a CNN trained on grayscale face crops in order to identify stress-revealing micro-expressions. Eye interaction was assessed via a CNN applied to eye-region crops, with forward facing and reduced blink rates indicating higher confidence. Hand movements were assessed via either CNN-based image classification or Transformer-based temporal modeling of hand key points, identifying confident vs. uncertain non-verbal activity.

B.1 Speech: Emotion Recognition for Confidence

Speech sounds were measured as a core index of confidence, derived from the documented role of prosodic variation such as pitch, intensity, and rhythm in differentiating confident from hesitant speech. Emotional valence and activation were also considered to be core correlates of confidence. To extract features, the 40-dimensional Mel-Frequency Cepstral Coefficients (MFCCs) and their first- and second-order derivatives were employed, optionally supplemented by spectral features such as roll-off, centroid and zero-crossing rate. The MFCC spectrograms extracted were used as input to an emotion classification CNN. The model consisted of three or four progressively larger filter sizes convolutional blocks [128, 256, 512], each of which included 3×3 kernels, ReLU activation, Batch Normalization and 2×2 max pooling layers. Dropout with a probability of 0.3 to 0.5 was applied after each block for regularization purposes. The output was flattened and passed to a dense layer of 256 units before the SoftMax classification final layer. Emotions were classified as positive (Happy, Neutral) and negative (Angry, Fear, Sad, Disgust, Surprise) classes as confident & non confident. Temporal scoring was employed, in which the ratio of positive emotion time to total time was used to find the speech emotion score. The score was also equally mapped to confidence features; each assigned a contribution by the emotion score.

B.2 Facial Expressions: Emotion Recognition for Stress

Face info features were utilized as a basic modality for stress recognition utilizing the fact that micro-expressions such as furrowed brows, lips being clenched and a stiff jaw are highly associated with psychological stress. Positive effect, however, was taken to represent decreased stress. Preprocessing was done in detecting and cropping the face from video frames and converting them to grayscale, resizing them to 48×48 pixels, and normalization. A CNN architecture was then used for feature extraction and classification. The model had four convolutional blocks with filter sizes [128, 256, 512, 512], each with 3×3 filters, ReLU activations, Batch Normalization, 2×2 max pooling, and dropout (0.3–0.5). The features were fed into a dense 512-unit layer followed by a SoftMax output layer. Emotions were defined as positive (Happy, Neutral) and negative (Angry, Fear, Sad, Disgust, Surprise). Facial emotion score was determined as the ratio of positive emotion time to the total time. This score played a stronger role in the representation of the stress-dominant role of facial analysis for the entire approach.

B.3 Eye Engagement: Confidence from Gaze Behavior

Eye contact was experimentally tested as a confidence proxy by assessing gaze direction, fixation patterns, and blink frequencies. Sustained forward gaze and stable fixation were viewed as confidence indicators, while constant gaze avoidance or fast blinking signaled uncertainty. The detections were coded as positive (forward gaze, consistent eye contact) and negative (gaze aversion, downward looking, or fast blinking) categories. Upon these detections, an engagement score was calculated as a proportion of positive eye behavior to overall time. For labeling engagement levels, eye regions were cropped and preprocessed to 48×48 grayscale images prior to being fed into a CNN. The model had several convolutional blocks with filter sizes [128, 256, 512, 512], followed by max pooling and dropout for regularization. The characteristics acquired were passed through a dense layer of 256 units before the output SoftMax layer. Since gaze behavior was assumed to be more reflective of confidence than stress, engagement score was mapped onto confidence B with additional weight.

B.4 Hand Gestures: Confidence from Non-verbal Movement

Hand gestures were a key modality for measuring confidence using non-verbal movement signals. Positive actions such as open palms, pointed gestures, and smooth co-speech gestures were associated with confidence, and increased confidence was observed when these gestures were observed, while negative actions such as fidgeting, touching, hiding, or random movement were associated with nervousness. Gesture scores were determined as the time spent on positive gestures divided by total time. Two simultaneous feature extraction approaches were explored: (i) a sequence point-sequence path, where 2D hand joint positions over time were sampled and input to a temporal CNN or Transformer encoder to learn motion dynamics; and (ii) an image-based path, where cropped hand or upper-body frames were identified from a CNN. For the Transformer-based sequence model, the significant point sequences of size (T×J×2) were fed into positional encoding layers and 2–4 Transformer encoder blocks with 4–8 attention heads and then a dense layer and SoftMax classifier. Since gestures were considered to convey confidence rather than stress, the final mapping placed high emphasis on confidence.

C. Training Protocol (All Subsystems)

For training, datasets were split into train, validation, and test sets in a subject-exclusive manner to prevent identity leakage. Optimization was performed with the Adam optimizer and categorical cross-entropy loss, while data augmentation (noise injection, brightness jitter, and random flips) improved robustness. Regularization included dropout and L2 weight penalties

D.Real-Time Prediction Pipeline

The real-time interview pipeline integrates audio and video streams via WebRTC and processes them through the trained models. Sliding windows (2–3 seconds for audio and 16–32 frames for video) enable continuous inference with minimal latency. Predictions are smoothed using an exponential moving average to avoid output flicker. Model serving is deployed using FastAPI endpoints with TensorFlow/PyTorch backends, while OpenCV handles real-time frame processing. Subsystem outputs are normalized to 0–100 confidence and stress scores

E. Score Fusion and Role-Aware Aggregation

The outputs from all four subsystems are aggregated into final confidence and non-stress scores. A weighted fusion scheme is employed, where each modality contributes proportionally based on its reliability: speech (split for confidence and non-confident), facial expressions (stress-dominant), eye engagement (confident and non-confident), and hand gestures (confidence-dominant). The fused scores are combined into a Final Emotional Fitness Score into a board using recruiter-defined role weights, allowing HR managers to adapt the evaluation depending on the job profile (e.g. customer support emphasizes stress management, while sales roles emphasize confidence). Candidates are then ranked according to their final scores.

Job Role	WnsW_{ns}	WcW_c
Sales Executive	0.30	0.70
Customer Support	0.60	0.40
Marketing Analyst	0.40	0.60

Table2: Role weight examples (editable by HR).

Illustration (Sales Executive):
Given NS=50, C=80, Wns=0.30W_ns} =0.30, Wc=0.70W_c=0.70:
Final Score= (50×0.30) +(80×0.70) =15+56=71\ Final Score} = (50 \times 0.30) + (80 \times 0.70) = 15 + 56 = 71

Ranking. Candidates are **sorted by Final Scores**; ties can be broken by higher Confidence or recruiter-chosen tiebreakers.

F. Evaluation Protocol and Metrics

The system was tested on labeled datasets specific to each modality. Baseline models were classical machine learning classifiers, including SVMs and shallow CNNs, for comparison against the proposed deep learning architectures. The evaluation metrics included accuracy, precision, recall, F1-score, confusion matrices, and inference latency. Ablation studies examined the contribution of each subsystem, the effects of multi-modal fusion, and the sensitivity to mapping coefficients. The statistical significance tests used McNemar's test and there were bootstrapped confidence intervals.

G.Implementation Details

The system was constructed with a focus primarily on the use of Python 3.x and using deep learning frameworks like TensorFlow or PyTorch, as well as supporting libraries that included OpenCV for video stream processing and librosa for retrieving audio features. We also provided a FastAPI backend that supported real-time inference and connection with front-end applications and provided optional deployment optimization through ONNX Runtime and TensorRT. Training and inference happened on GPUs enabled with CUDA but could also be used with CPU since the architecture supports both for deployment in restricted environments. Configuration management was achieved through YAML and JSON files for storing hyperparameters, training settings, and other files for the configuration management, as well as using a model registry for versioning and keeping proper track of the trained models. For monitoring, the system kept inference logs for tracking latency and FPS performance and facilitated statistical checks for feature drift verifying that the data distribution shifted over time. Finally, we stored anonymized audit trails to protect the accountability and transparency processes of the licensee.

H. Reliability, Fairness, and Privacy Controls

To maintain methodological robustness and equity, cross-subject validation was utilized during training to reduce identity leakage and to enhance generalizability. Also, wherever data splits were available, they were stratified by gender and age, to enable comparisons in error rates across demographic strata to identify potential sources of bias. Calibration methods were used whenever appropriate, including when binary classification thresholds were required for content and features, to derive well-calibrated confidence scores. As part of fairness checks, performance gaps across demographic groups were assessed, when information was legally and ethically permitted. From a privacy perspective, run as a multiteam system, the system itself ensured that processing of all multimodal streams always occurred locally, apart from when the end-user voluntarily submitted data for storage, and altogether avoided the storage of recorded videos or audio data of any kind without explicit consent of the End User. In situations where retention of discrete data was required, this was stored encrypted at rest and meeting relevant security measures, and End Users were made aware of data retention policies, which wherever possible contained details regarding opting out.

I. Limitations and Risk Mitigations (Methodological)

The proposed application has some methodological limitations, and associated mitigation strategies were implemented to address them. One major area of concern was environmental sensitivity since noisy backgrounds, poor lighting situations and variations in cameras can lead to extremely unreliable captured signals. Risk was mitigated by utilizing data augmentation techniques, denoising filters and robustness testing to check for consistency in a variety of environments while building the training pipeline. Another limitation relates to accessories and occlusions, such as glasses, face masks or hand obstructions which can impact accurate detection. The advantages of multimodal fusion are realized: when a signal is unreliable, other modalities may be reliable, and cued behavior can be compensated for. Another methodological limitation involved cultural or individual specificity of expressive behavior, which may differ from the expected shape or may result in a non-typical expression altogether, which could lead it to be misinterpreted. The system had configurable weighting controls, allowing HR decision-makers an opportunity to make the system custom to their preferences. We discouraged the idea of having an automated prediction be a final candidate assessment and emphasized that this relatively low-stakes intervention would involve human-in-the-loop oversight. This ensured that the final candidate assessment would not be solely automated.

J. Reproducibility

Reproducibility was evaluated as a function of properly documenting the methodology to make results reliably replicable. To control the randomness from libraries like NumPy, TensorFlow, and PyTorch to produce experimental consistency, we fixed random seeds used across libraries. All datasets and trained model artifacts were versioned. Built scripted pipelines to rationally cover the whole of training, evaluation, and deployment. Hyperparameters and experimental configuration were stored in fixed configuration files to enable reproducibility and reduce manual intervention to replicate an experiment. We also recorded the exact data set splits that were used in experiments, so that evaluation still held comparability moving forward. Collectively, these reasonable measures allowed for a consistent and transparent base to establish repeatability across multiple trials.

The output of the approach can be demonstrated by way of an example candidate ranking for a sales executive role. Each candidate is rated on two traits: non-stress (NS) and confidence (C), calculated from the four modalities with pre-defined weights and each of these scores are calculated into a scoring dashboard. In this example, candidate A scored 60 in NS and 80 in C, candidate B scored 50 in NS and 90 in C, and candidate C scored 70 in both NS and C. Based on these we selected the best candidate to fill the role. Our example demonstrates how the approach integrates multimodal signals into interpretable, weighted scores that can be modified as the result of human resources policies.

Candidate	Non-Stress (NS)	Confidence (C)
A	60	80
B	50	90
C	70	70

Table3: Example: Candidate Ranking (Sales Executive)

IV. Results and Discussion

A. Statistical Performance Overview

The proposed AI-based candidate recommendation was evaluated across its four core modules: speech emotion analysis, facial expression recognition, eye engagement detection, and hand gesture recognition. Each subsystem was trained and validated on its respective datasets using convolutional neural networks (CNNs). For benchmarking, classical machine learning models such as Random Forests and Support Vector Machines (SVMs) were also implemented as baselines. Table IV presents a summary of the performance across components.

Component	Model Used	Accuracy	F1 Score
Speech Emotion Detection (Confidence)	CNN	91.4%	0.89
Facial Expression Analysis (Stress)	CNN	88.7%	0.86
Eye Engagement Detection (Confidence)	CNN	85.2%	0.84
Hand Gesture Recognition (Confidence)	CNN + Random Forest	87.6%	0.86

Table4. Component-Wise Performance

The findings indicate that all four modalities produced accuracy rates over 85%. This confirms the validity of multimodal behavioral analysis for predicting stress and confidence. Speech emotion detection had the highest accuracy level at 91.4% and an F1-score of 0.89 among the subsystems. This indicates great predictive value for acoustic features - pitch, duration, energy, etc. - for indicating confidence. Facial expression analysis also performed well indicating that micro-expressions and facial tension are reliable indicators of stress.

B. Interpretation of Results

The results show that confidence detection can be inferred based on many verbal and non-verbal modalities. All components of speech, eye engagement, and hand gestures were important supplementary cues. These findings corroborate with previous studies that demonstrate that gaze direction, fewer blinks, and intentional hand movements correlate with behavior that signifies confidence. The stress detection results, which relied primarily on the analysis of facial expressions, also conformed to pre-existing psychological and behavioral theories, which signify that stress is externally identifiable based on facial muscle tension and micro-expressions. It is worth noting that the multimodal fusion approach was stable in real-world conditions as when one input (i.e., eye tracking) was reduced due to glare, glasses, or excess blinks et al., the other modalities were still reliable. This overlap would solidify how multimodal systems are superior to unimodal approaches, which have been stated in previous studies.

C. Novel Contributions

The results highlight several unique contributions of this work compared to previous literature. First, this approach highlights the real-time candidate evaluation with a unique blend of four behavioral modalities: speech, facial expressions, hand gestures and eye engagement. Second, the user specified role weighting allows recruiters to modify the relevance of the evaluations by giving more importance to confidence when evaluating sales executive candidates for example or applying more importance to stress management for customer service roles. Finally, unlike existing unimodal or experimental approaches we showed a unique real-time analysis that enables a ranking of candidates for each role in real-time in an "on-the-job" sound shown above, and or interviews (screening) that are currently impractical (due to the need to interrupt and record the interviews).

D. Limitations

While the results are promising, some methodological limitations exist. Speech analysis accuracy deteriorated in noisy surroundings or with poor microphone quality and facial and hand gesture recognition was sensitive to individual variations in lighting and camera resolution. Eye engagement analysis was less accurate for glasses and for people exhibiting abnormal blinking styles. Other potential limitations could stem from the role of cultural variation in gaze behavior and gesture interpretation, and participant generalizability from a larger pool. Furthermore, the approach currently assesses only confidence and stress from an extremely narrow range of psychological constructs without broader personality types, such as the Big Five traits, that might add an interesting dimension to the candidate analysis.

E. Implications and Future Work

The results have shown the twofold potential of multimodal AI to positively affect hiring processes by providing objective and data-driven measures of confidence and stress, while facilitating a reduction in subjectivity and bias in evaluating candidates. Future studies should be directed toward expanding neural networks (inputs) by increasing the diversity of datasets available to optimize cultural and demographic generalizability, improving robustness through environmental challenges (e.g. sound, light), and increasing additional behavioral markers (e.g. various postures, and speech semantics). In addition, extending the context of the approach into multiple languages and cultural settings would increase applicability. Personality trait evaluations would allow for much more extensive candidate profile evaluations and a greater potential for accuracy of matching appropriate staff to position roles.

V. CONCLUTION

The AI recruitment application employs multimodal analysis speech, facial, hand, and eye interaction to forecast confidence and stress levels with more than 85% accuracy. It allows real-time, role-based assessments, increasing fairness and transparency. In spite of constraints from environmental and cultural factors, it serves as an example of how AI eliminates subjectivity in soft-skill measurement.

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